

Part V

Conformal Prediction

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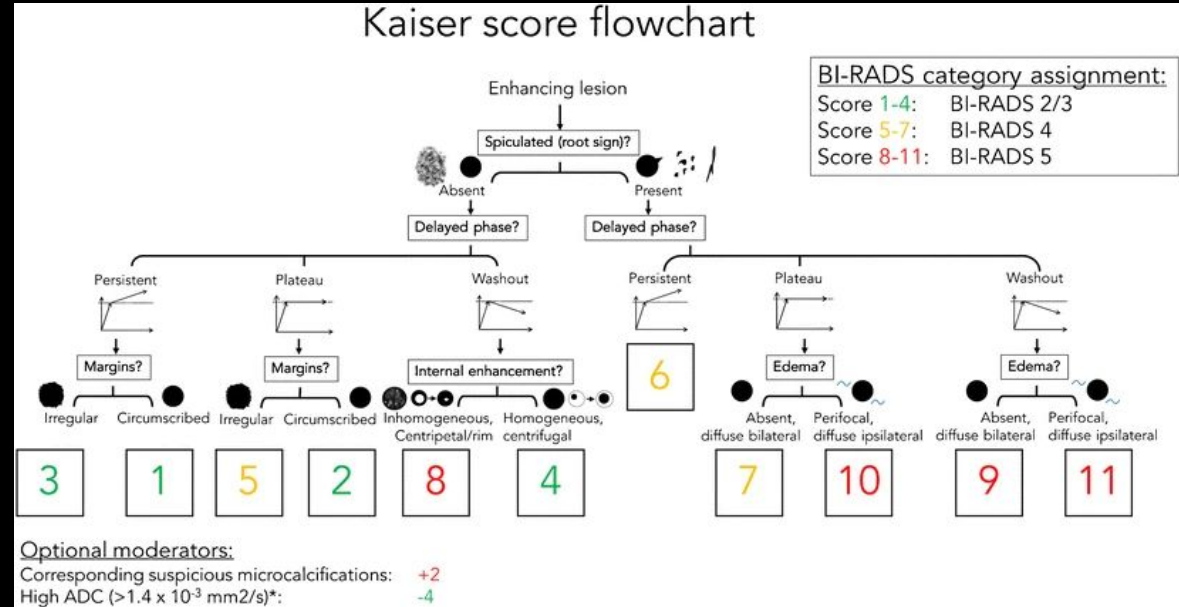
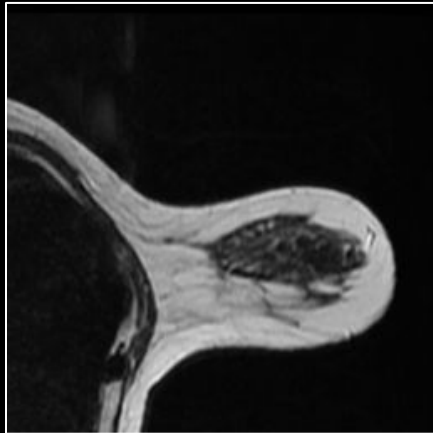
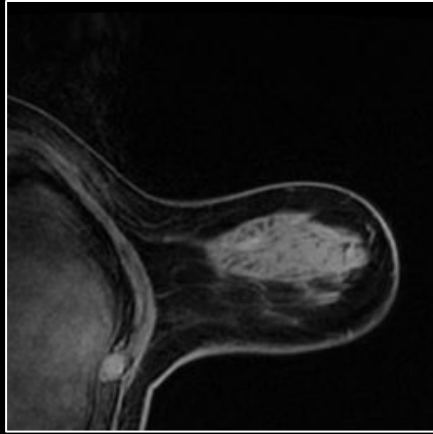


Conformal Prediction - Contents

1. Motivation
2. Conformal Prediction: Ingredients
3. Conformal Predictions: Algorithm
4. Hands-On

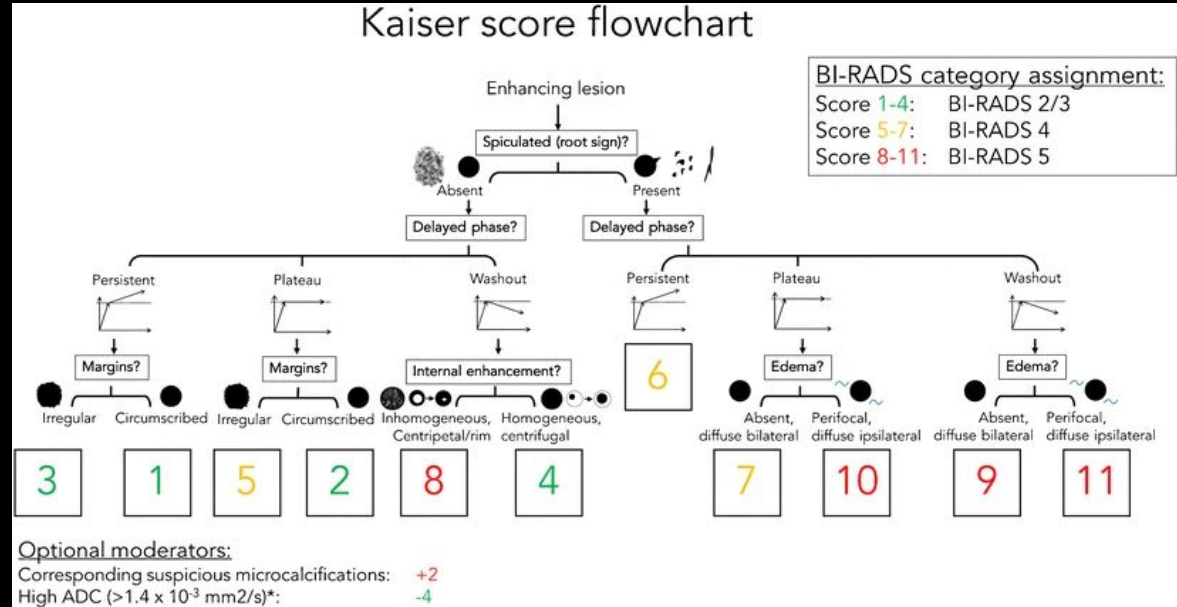
1. Motivation

Dietzel M, Baltzer PAT. How to use the Kaiser score as a clinical decision rule for diagnosis in multiparametric breast MRI: a pictorial essay. Insights Imaging. 2018



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1. Motivation: Differential Diagnosis

Step 1: Do we see an enhancing lesion?

- On MRI, when you inject contrast, suspicious lesions often “light up” because they have abnormal blood vessels.
- So the first question is: *does the lesion actually enhance?* If yes → move on.

Step 2: Is there a spiculated margin (“root sign”)?

- **Spiculation** means the lesion has spikes or radiating lines extending into the surrounding tissue — like roots of a tree.
- This is a strong red flag for cancer. If it’s present, you follow the *right-hand branch*.
- If absent (the lesion looks smooth/rounded), you go to the *left-hand branch*.

Step 3: What happens in the delayed phase (contrast wash-out curve)?

This is the part you asked about — it’s the **time course** of contrast enhancement.

- After contrast injection, radiologists watch how bright the lesion gets over time.
- There are three typical “curves”:
 - **Persistent:** keeps getting brighter and brighter with time.
→ Usually benign (think of a sponge slowly soaking water).
 - **Plateau:** gets bright quickly, then levels off.
→ Suspicious (like tissue that soaks fast but then “caps out”).
 - **Washout:** gets bright early, but then *fades* as contrast drains away.
→ Very suspicious for cancer (because malignant tumors often have “leaky” vessels).

So the “**delayed phase**” check is: which of these three time-curves does the lesion follow?

Step 4: If persistent or plateau → check margins

- **Margins** = edge of the lesion.
- Smooth (circumscribed) edges → usually benign.
- Irregular/jagged edges → more worrisome.

Step 5: If washout → check internal enhancement pattern

- Inside the lesion, how does the contrast distribute?
- If it’s patchy, rim-shaped, or irregular → higher suspicion.
- If it’s uniform or “centrifugal” (from inside out), less worrisome.

Step 6: If spiculation was present (right side of the tree) → check delayed phase again, then edema

- If the lesion is spiculated and the curve is persistent/plateau/washout, you still branch down.
- In later branches, radiologists look for **edema** — swelling in the tissue around the lesion.
- Edema often shows up in malignancy, so its presence increases the suspicion.

Putting it together

At the bottom of the tree, you land on a **Kaiser score number (1–11)**.

- **Low scores (1–4):** likely benign (BI-RADS 2/3).
- **Middle scores (5–7):** indeterminate but suspicious (BI-RADS 4).
- **High scores (8–11):** very suspicious, likely malignant (BI-RADS 5).

2. Conformal Prediction: Ingredients

Suppose you've got an FDA-approved diagnostic model

$$\hat{\mathcal{M}}_y(x) \sim \mathbb{P}(Y = y | X = x) \quad y \in \{1, \dots, K\} = \mathcal{Y}$$

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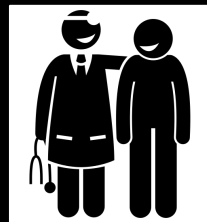
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with $p = 0.9$



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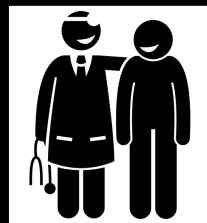
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with $p = 0.9$



Predict a set $\mathcal{T}_{x^*} \subseteq \mathcal{Y}$ which contains y^* *with high p*

$$\text{Coverage : } \mathbb{P}(y^* \in \mathcal{T}_{x^*}) \geq 1 - \alpha$$

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Prediction Sets : $x^* \mapsto \mathcal{T}_{x^*} \subseteq \mathcal{Y} = \{1, \dots, \mathcal{K}\}$

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{ fox
squirrel
0.99 }



{ fox, gray, bucket, rain
squirrel, fox, barrel
0.82, 0.03, 0.02, 0.02 }



{ marmot, fox, mink, weasel, beaver, polecat
0.30, 0.22, 0.18, 0.16, 0.03, 0.01 }

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No retraining allowed, but you have some fresh data available:

$$(\text{bMRI}_1, \mathbf{y}_1), \dots, (\text{bMRI}_N, \mathbf{y}_i) = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N \sim \mathbb{P} \text{ i.i.d.}$$

We call this a Calibration Set

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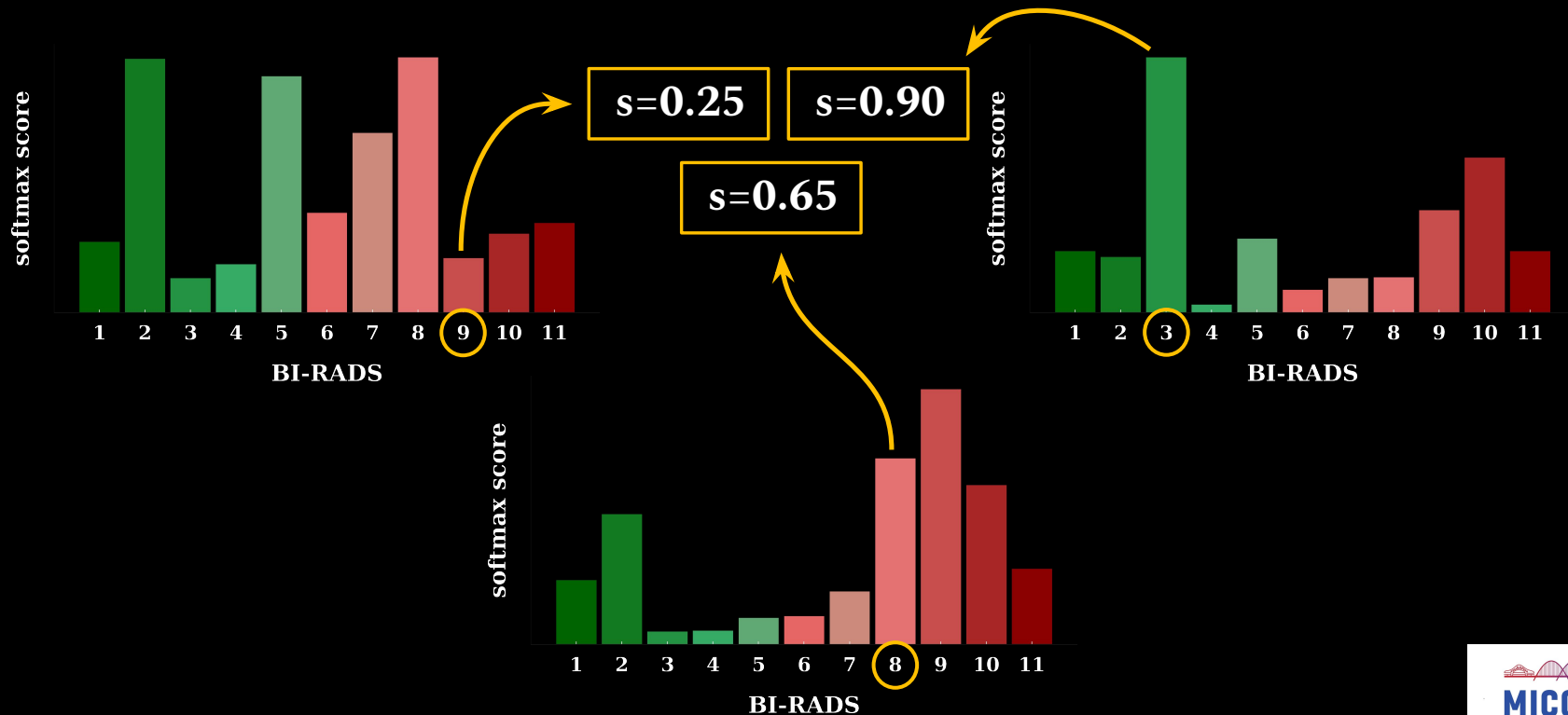
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How do we build these Conformal Prediction Sets?

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$E=\{0.25, 0.30, 0.40, 0.50, 0.60, 0.65, 0.70, 0.75, 0.75, 0.90\}$

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$E=\{0.25, | 0.35, 0.40, 0.50, 0.60, 0.65, 0.70, 0.75, 0.75, 0.90\}$

$t=0.35$ -> 90% of true-class scores are $\geq t$

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4. Use **$t=0.35$** to pick returned categories on a test sample

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$$s=[0.25, \mathbf{0.50}, 0.25] \quad T=\{\mathbf{C1}\}$$

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$$s=[0.20, \mathbf{0.40}, \mathbf{0.40}] \quad T=\{\mathbf{C1}, \mathbf{C2}\}$$

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2. Collect all scores $E=\{0.25, 0.90, \dots, 0.65\}$
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$$s=[0.33, 0.33, 0.34] \quad T=\emptyset$$

3. Conformal Prediction: Algorithm

For a desired miscoverage of α , find a value $\hat{q}_\alpha$ such that we keep $1-\alpha$ of the scores in E :

$$\hat{q}_\alpha = \text{np.quantile}([E_1, E_2, \dots, E_N], \alpha)$$

There is a finite-sample correction that enables the conformal coverage guarantee; it grows as the calibration set gets smaller.

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If we use $\hat{q}_\alpha$ to build prediction sets on test data, we have theoretically guaranteed coverage:

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if data is interchangeable.

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4. What would be next

This was Split Conformal Prediction. Extensions:

- Better prediction sets: **APS, RAPS**
- Beyond marginal coverage: **conditional coverage/**
- Beyond classification: **regression, segmentation**
- Beyond coverage: **Conformal Risk Control**

4. Hands-On

Github repository:

<https://github.com/agaldran/uqinmia-miccai>

